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Domain: Testing

Azure Fundamentals

Scenario 1:

Your team needs to deploy a virtual machine in Azure to test a new software application.

Here the team has requested both windows and Linux virtual machine.

Question:

How could you set up these virtual machines and what considerations are needed for pricing and OS licensing?

Setting up a Virtual Machine depends on the platform we are using like AWS,Azure or Google Cloud.

Step 1: Sign in to Azure Portal with Microsoft account

Step 2: Create a new Virtual Machine

Step 3: Select Azure Subscription in which VM will be created. Create a new existing group that hold related azure resources

Step 4: Provide several details about VM. (Name, Region of VM will be hosted, Availability options,Image,Size,Authentication Type,Username>Password or SSH Key)

Name:NaveenyaVM

Region of hosting: West Europe

Image: For windows server: Windows server 2022

For Linux: Ubuntu

Size:1 vCPU,1GB RAM

Authentication Type: SSH public key for Linux VMs,Password for Windows and Linux VMs

Step 5: Selecting an existing virtual network or create a new one. Choose an existing subnet or create a new VM. A public IP will be assigned to VM

Step 6: Select the type of OS disk (SSD or HDD) SSD are faster but higher in cost. Additional data disks can be attached for extra storage.

Step 7: Review all the settings, once confirmed click Create

Step 8: Once Virtual Machine is deployed, we can be able to connect

For Windows VM:Connect via Remote Desktop Protocol
(RDP)

For Linux VM:Connct via SSH

Scenario 2:

The IT security team has requested that sensitive data stored in azure storage account be encrypted to meet compliance requirements.

Question:

How could you ensure that data stored in azure storage is encrypted, and what encryption types are available?

Answer:

Azure provides robust encryption mechanisms for both data at rest and data in transit (data being transferred).

Encryption Available: i)Data Encryption at Rest

Azure Storage automatically encrypts data at rest to help meet compliance and security requirement

- A) Storage Service Encryption (SSE)
- B) Encryption Key Management Options
- C) Azure Blob Storage Encryption Options
- D) Azure Disk Encryption

li)Data Encryption in Transit

Azure ensures that data in transit is encrypted using protocols like TLS (Transport Layer Security).

- A) TLS Encryption for Data in Transit

lii) Encryption Types Available in Azure Storage

- A) Storage Service Encryption (SSE)
- B) Client-Side Encryption
- C) Azure Disk Encryption (ADE)
- D) Encryption at the Application Level

Scenario 3:

You are responsible for setting up devops pipeline in azure devops for your application. The pipeline must deploy code to an azure app service and notify the team if the deployment fails.

Question:

How could you configure this pipeline to meet the requirements?

Answer:

To set up a DevOps pipeline in Azure DevOps that deploys code to an Azure App Service and notifies the team if the deployment fails.

Step1: Make sure you have an Azure DevOps organization, and a project set up

Step2: Navigate to Azure DevOps Portal

Step3: Focus on setting up a YAML pipeline, which gives more flexibility and is widely used for CI/CD in DevOps.

Step4: Need to build your application (e.g., .NET, Node.js, or Java-based app)

Step5: After building the app, need to deploy it to an Azure App Service

Step6: To notify your team in case the deployment fails, you can integrate Azure DevOps notifications or use a third-party service like Microsoft Teams.

Scenario 4: Your organization is moving its on-premises data to azure. The data base must remain accessible during migration with minimal down time

Question:

Which azure service could you use, and how could you perform the migration?

Answer:

When moving an on-premises database to Azure with minimal downtime, the goal is to ensure that the database remains accessible during migration and that the process is as seamless as possible. One of the most suitable Azure services for this kind of migration is Azure Database Migration Service (DMS)

Step1: Before starting, use Azure Migrate or Data Migration Assistant (DMA) to assess your on-premises database.

Step2: Set Up Azure Database Migration Service

Step3: Prepare the Source Database for Migration

Step4: Select the Migration Method

Step5: In Azure DMS, you will need to configure the source and target databases

Step6: Once the migration starts, you can monitor the progress via the Azure portal or DMS dashboards.

Step7: Once the data is fully replicated and synced to Azure, there will be a **cutover** window where the last few changes are replicated

Step8: After the migration is complete, perform post-migration checks to ensure everything is working as expected.

